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Paleomagnetic results from Permian rocks of the northern Saharan craton and motions of the Moroccan Meseta and Pangea

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The characteristic magnetization of redbed samples from the upper part of the Série d'Abadla (probably Early Permian 31°N, 2.7°W) has a mean direction derived from 13 sites of $D=129°$, $I=11°$, $k=59$, $\alpha_{95}=6°$ and a corresponding south paleopole at 29°S, 60°E, $A_{95}=5°$. All directions have reversed polarity. The paleolatitude of the northern fringe of the Saharan craton was 6° ± 5°S, which is in excellent agreement with that for the Moroccan Meseta. Therefore, in all probability, there has been no paleolatitudinal displacement greater than about 500 km of the Moroccan Meseta relative to Africa since Permian time. Comparison of results from sedimentary rocks shows no evidence for relative rotation of the Moroccan Meseta since Permian time. Small apparent rotations are indicated by evidence from massive trachyandesite lavas from Morocco, but we argue that these could have arisen from the incomplete averaging of secular variation and uncertainties in estimates of paleohorizontal, rather than from true tectonic rotations. The combined latest Carboniferous/Early Permian paleopole for the Saharan craton and the Meseta differs form the path of apparent polar wandering for North America when the continents are assembled in Wegener's Pangea (Pangea *A*, in which northwest Africa is opposite North America). It is in reasonable agreement when the continents are assembled in the Pangea *B* configuration (northwest Africa opposite Europe).

1. Introduction

It has recently been shown that there was no major movement between the Moroccan Meseta and the remainder of Africa during and since the early phases of the opening of the Atlantic Ocean (approximately 180 m.y.), that is since the Early Jurassic [31]. It is of further importance to determine if any earlier relative motions have occurred, to assist in understanding the earlier geological history of the region, and in describing global continental drift in the Triassic and Late Paleozoic (approximately 300 to 200 m.y.). For the later problem Morocco and northwest Africa are important because they were centrally placed in Pangea; they are key elements in the Late Paleozoic jig-saw puzzle.

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Numerous paleomagnetic results are available from Late Paleozoic and Mesozoic rocks of the Moroccan Meseta, and general comparisons have been made [1–7] with results from distant parts of Africa and Gondwana. But for detailed testing it is important to obtain results from the northern Saharan craton immediately to the south Atlas Fault. We present results that now allow such comparisons to be commenced.

2. Geology

The Abadla Basin is situated near Bechar, close to the northern edge of the Saharan craton (Fig. 1). The Série d'Abadla overlies a Westphalian sequence [8] containing coal (Fig. 2). It is cut by dolerite dykes, overlain by basalts and capped by Neogene sedimentary rocks of the Hamada For-

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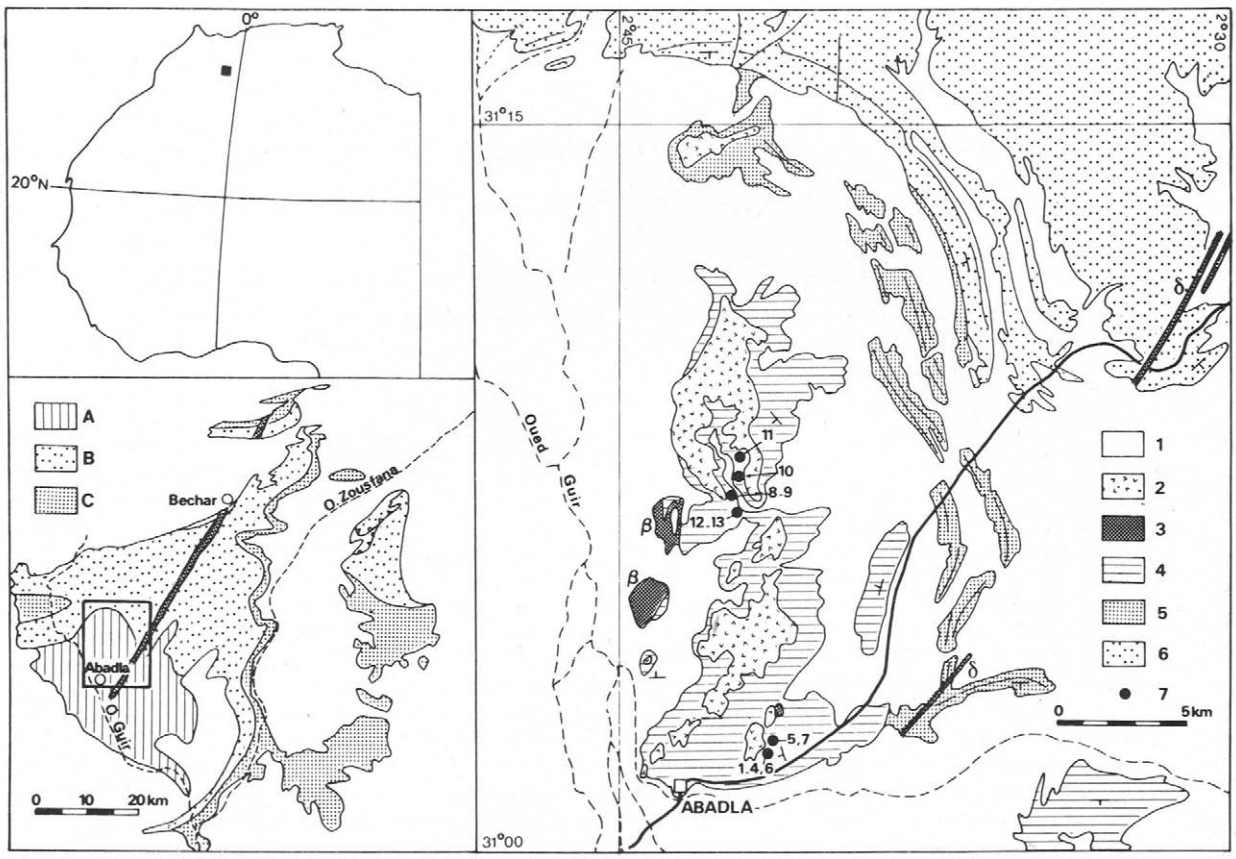


Fig. 1. General geology is shown bottom left: *A* = Série d'Abadla; *B* = Upper Carboniferous; *C* = Lower Carboniferous. More detail is given on the right; *1* = Quaternary; *2* = Hammada (Neogene); *3* = doleritic basalts (flows β and dykes δ); *4* = upper shales and sandstones, and *5* = lower calcareous shales of the Série d'Abadla; *6* = coal measures (Westphalian); *7* = sampling sites.

mation. The volcanics have not been dated in this area, but to the west, in Morocco and further south, similar igneous rocks have yielded Early Jurassic radiometric ages [9,10]. The Série d'Abadla [11] consists of tow main units—a lower sequence of shales with intercalated sandstones and limestones, and an upper redbed unit. The sequence is conformable throughout and the dips vary from 8 to 10° to the west. Limestones at the base contain lacustrine stromatolites probably of Stephanian age [12]. Floras collected from the middle of the lower unit indicate an Autunian, but do not exclude a Stephanian age (Doubinger and Fabre, personal communication). The upper redbed unit has yielded no fossils. It is conformable downward with the lower unit indicating that it is Permian, and most likely, Early Permian. There are no Triassic rocks known in the immediate area, although Late Triassic beds occur in Morocco, north of the South Atlas Fault.

The collections were made in February 1979. The Série d'Abadla totals about 2000 m in thickness and the sampling sites span about 450 m spaced over two localities about 10 km apart (Figs. 1 and 2). Because of their suitability for magnetic studies the redbeds have been sampled. They consist of shales, muddy siltstones, sandstones and occasionally conglomerates. The shales are too friable to sample. All samples were obtained from muddy siltstones, and fine or very fine sandstones. Balls of hard impervious muddy siltstones occur within the shaly beds. These were

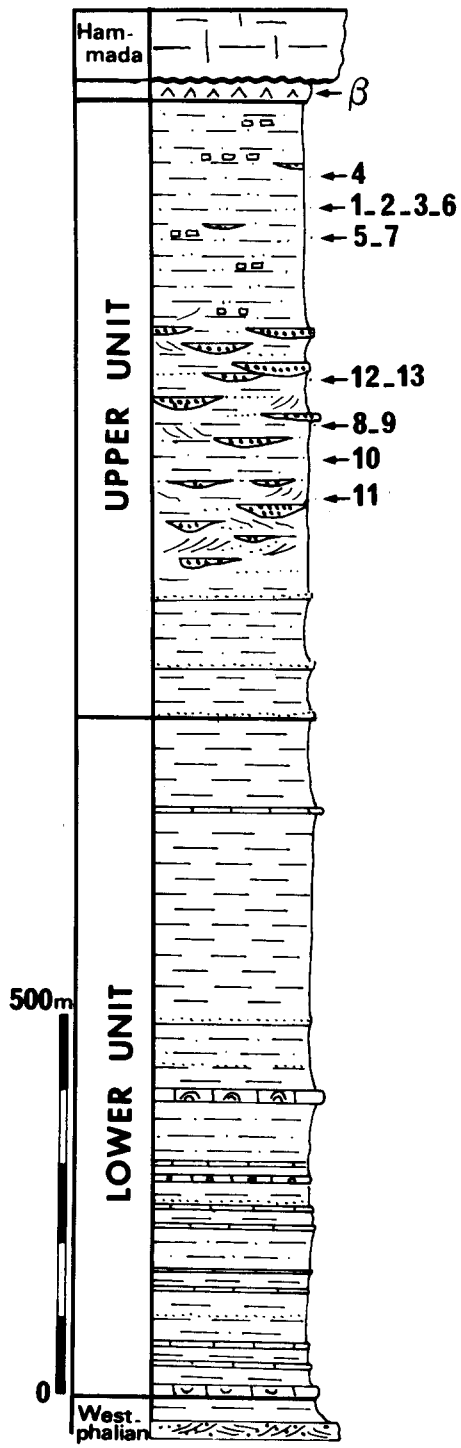


Fig. 2. Stratigraphic column of the Série d'Abadla. Sampling levels are numbered. Flows of unit 3 of Fig. 1 are labelled "β".

collected wherever possible and give excellent results. The shales and the matrix to the muddy siltstones and sandstones are typically a mud-grade mixture of illite, chlorite and kaolinite. Gypsum is common. The sandstones are commonly cross-bedded. The beds are typical stream-channel deposits. The source-area appears to have been to the west. Evidently these beds were laid down in a hot continental climate with alternating wet and dry seasons.

3. Results

Oriented samples were collected from 13 sites either by drilling or by hand sampling of softer rocks using plaster or Paris caps [28] for orientation (Figs. 1 and 2). All samples were from the upper shale-sandstone unit (Fig. 2). The muddy siltstones have negligible porosity. The sandstones are solid but are slightly permeable. One to three specimens were cut from each sample, altogether 48 oriented samples and 95 specimens.

The directions of natural remanent magnetization (NRM) are directed toward the southeast far from the present field (Fig. 3). The muddy siltstones all have shallow inclination generally about 10°. Some sandstones also have shallow inclination, but most are intermediate ranging up to 50°.

Typical behaviour of muddy siltstone during thermal demagnetization is shown in Fig. 4. The decay of intensity is slow and there is an abrupt fall between 650 and 670°C. At 700°C the magnetization is weak and randomly oriented. The direction changes observed during treatment up to 670°C are negligible. Orthogonal vector diagrams [29] are linear with essentially perfect convergence to the origin. Therefore only one magnetization is present in the muddy siltstone. It has unblocking temperatures distributed over a range typical of that for hematite. We refer to this as the characteristic or *C* magnetization in the sense of Zijderveld [13].

The behaviour of sandstones (all are fine or very fine grained) is generally somewhat different from that of the muddy siltstones (Fig. 5). The NRM directions are steeper than those in the muddy siltstone, but after treatment at 300°C the directions are very similar. The remanence (S)

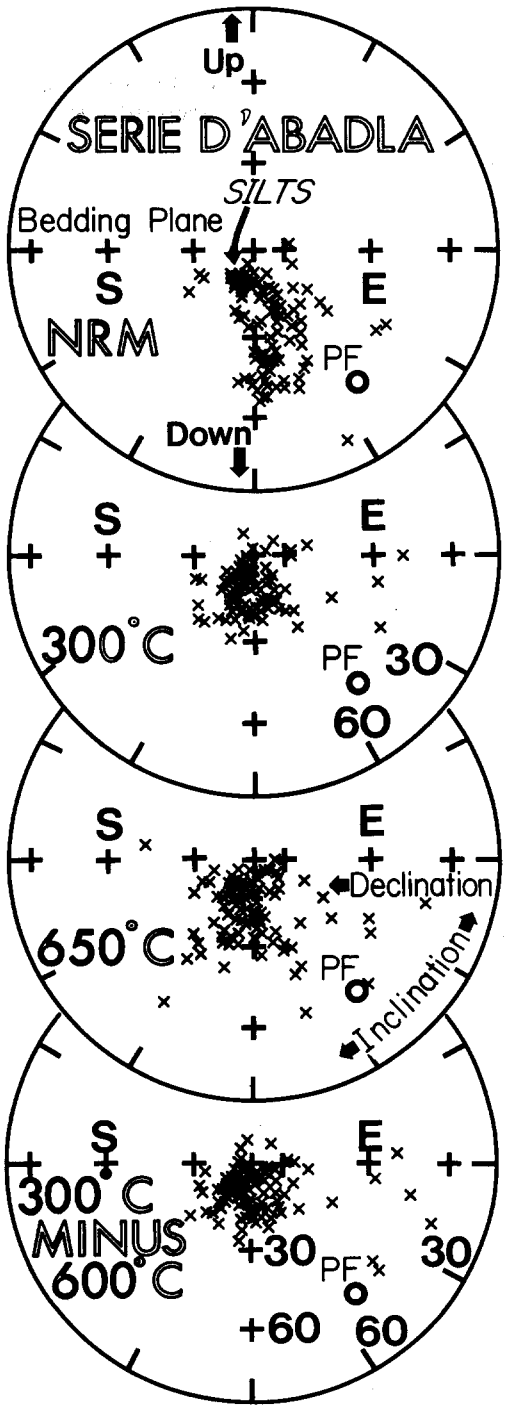


Fig. 3. Directions of magnetization plotted on upper hemisphere of an equal-area projection. The projection is arranged so the distribution of directions is centred approximately. The horizontal is the bedding plane at each collecting site and geographic south direction (S) points outward and to the reader's left. *PF* is the direction of the present earth's field (downward). At the top the NRM directions of all specimens are shown. In the middle the directions after cleaning at 300 and 650°C are given. Below the directions of the magnetization obtained by vector subtraction along the straight-line portions (300–600°C) of the orthogonal diagrams [29] are shown.

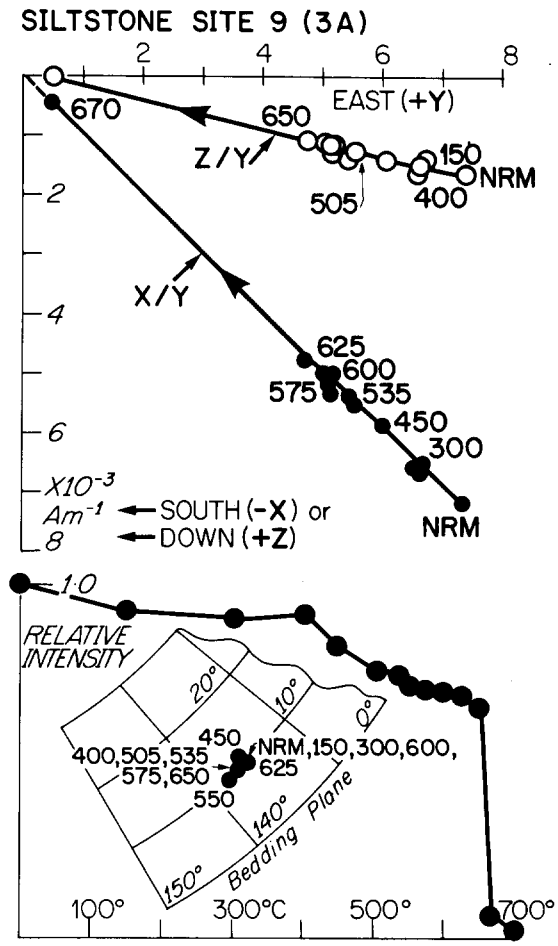
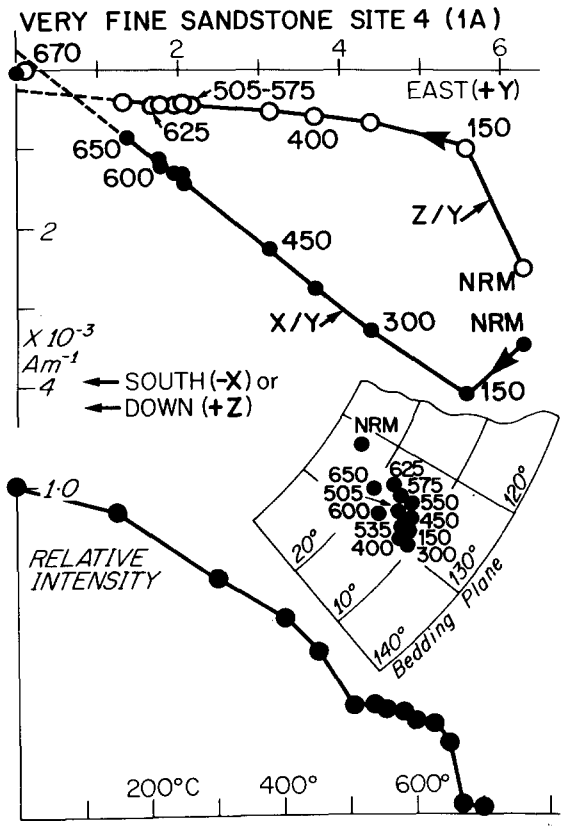


Fig. 4. Thermal demagnetization of red siltstone from site 9. Orthogonal diagrams [29] showing excellent convergence to the origin are shown above. Normalized decay curve (NRM intensity 104×10^{-3} A m^{-1}) and direction changes in the SE portion of an equal-area plot are shown below.

removed between 20 and 300°C is directed approximately along the present field. Between 300 and 650°C there is linear decay of the remanence not exactly toward the origin (Fig. 5). The remanence with unblocking temperatures between 300 and 670°C is therefore composed of two magnetizations, a larger magnetization similar in all essen-



.Fig. 5. Thermal demagnetization of very fine-grained red sandstone from site 4. Orthogonal diagrams [29] show good but imperfect convergence on the origin. Normalized decay curves (NRM intensity 8×10^{-3} A m^{-1}) and direction changes below.

tial respects (notably similar low inclinations) to the *C* magnetization of the muddy siltstones, and a much smaller magnetization with steeper inclination and a variable declination referred to as *H*. Although the unblocking spectra may overlap, that of *H* is generally higher than that of *C*. The effect of progressive thermal treatment is to increase slightly the inclination as the magnitude of *H* increases relative to *C*. But it is important to note that the effect is small because *H* is small and variable in direction from specimen to specimen; for example in Fig. 5 the inclination is 11° at 300°C and 20° at 650°C. The bias imposed on *C* by *H* can be effectively removed by vector subtraction as discussed below.

Eleven specimens have been demagnetized by alternating fields (AF) up to 280 mT [14]. The main effect is to remove the component *S*, but *C* and *H* could not be separated. Evidently the coercivities of *C* and *H* exceed 280 mT, as would be expected for single-domain hematite.

Eight specimens were subjected to chemical leaching [15,16] with unsatisfactory results. Five specimens completely disintegrated in 24 hours owing apparently to the muddy matrix and traces of gypsum. Three specimens (all fine sandstone) survived for 384 hours but disintegrated after that. Up to that time the vector diagrams show a roughly linear decline to the origin after the removal of *S*, but they are insufficiently well-defined to separate *C* and *H*.

We interpret *S* as a recent viscous remanent magnetization. The origin of *H* is uncertain. It might be attributable to anisotropy. It might be caused by the chemical formation of hematite (chemical remanent magnetization, CRM). The sandstones (*H* is found only in sandstones) although not highly porous, could have been repeatedly saturated with water during the rare rain showers which could have produced a CRM either recently or in the more remote past. *H* is absent in the muddy siltstones perhaps because they are impervious.

TABLE 1

Results from collecting sites of Fig. 1. The magnetization given is that removed in the interval 300 to 600°C

Site	*G*	Intensity	*D* (°), *I* (°)	*N* (*R*)
1	vf	8.57	129, 20	2 (1.99)
2	vf	7.39	137, 19	8 (7.92)
3	vf	6.80	117, 21	9 (8.49)
4	vf	6.97	111, 08	6 (5.59)
5	f	5.61	125, 06	7 (6.82)
6	vf, s	6.22	133, 16	8 (7.90)
7	f	3.62	134, 13	4 (3.98)
8	s	55.00	134, 10	10 (9.98)
9	s	81.20	139, 08	8 (7.96)
10	vf	8.93	123, 03	9 (8.89)
11	s	9.84	128, 03	8 (7.53)
12	s	12.70	141, 08	4 (3.95)
13	vf	6.73	124, 05	12 (11.40)

G is grain-size on Wentworth scale: s = siltstone; vf = very fine sandstone; f = fine sandstone. Mean intensity of NRM is in A m$^{-1}\times10^{-3}$. *D*, *I* are mean declination and inclination relative to bedding of the *C* magnetization. *N* and *R* are the number of specimens and the resultant.

TABLE 2

Summary results of thermal treatment

Treatment	R	D_m (°), I_m (°)	k	α_{95} (°)	λ (°S), φ (°E)	A_{95} (°)
1. NRM	12.54	125, 28	26	8	—	5
2. 300°C	12.81	130, 13	63	5	29, 58	5
3. 550°C	12.78	129, 16	54	6	28, 57	5
4. 650°C	12.74	130, 19	46	6	27, 55	5
5. 300–600°C	12.80	129, 11	59	6	29, 60	5

The mean giving 13 sites unit weight for three temperatures are given in entries 2, 3 and 4. Entry 5 gives the mean direction of magnetization removed in interval 300 to 600°C. D_m, I_m are the mean direction relative to bedding. R is the resultant, k the precision and α_{95} the radius of the circle of confidence [17]. The latitude (λ) and longitude (φ) and error A_{95} of the paleopole, given in the last three columns, are calculated by giving unit weight to each of the 13 site poles.

The C magnetization is present in all specimens, it is always the dominant magnetization and it persists with essentially consistent direction through 400 m stratigraphically (Table 1). For these reasons we regard C as the remanence acquired during formation of the Série d'Abadla. The presence of H reduces the accuracy with which the direction of C can be obtained. But its intensity is small and direction non-uniform, so that even after thermal treatment at 650°C (allowing H its maximum effect) the error introduced would not alter our conclusions. This can be seen in Table 2 in which the mean directions after treatment at 300, 550 and 650°C and those obtained by vector subtraction 600–300°C are given; the directions and paleopoles obtained are not significantly different. it is, however, desirable to remove H and therefore the most accurate definition of the field at the time of formation is, we believe, entry 5 of Table 2.

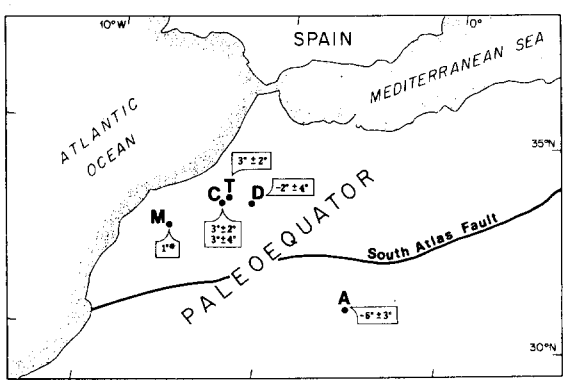


Fig. 6. Paleolatitudes from Morocco and Algeria compared. The paleolatitudes (negative is south latitude) and standard errors at five localities are shown. The localities are: M=Mechra ben Abou (18); c=Chougrane [18,19]; T=Taztot [18]; D=Djebel Tarhat [6]; A=Abadla (Table 2). The inferred position of the paleoequator is indicated.

4. Comparison with results from Morocco

Results are available from Permian rocks at four localities on the Moroccan Meseta (Fig. 6).

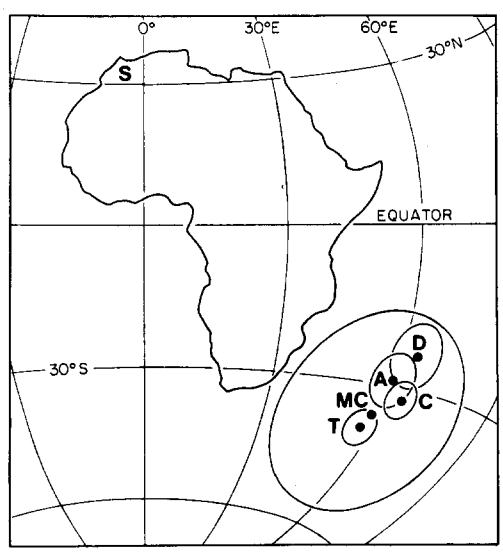


Fig. 7. Paleopoles from Morocco and Algeria compared. MC=Mechra ben Abou and Chougrane: rhyodacites and trachyandesites [18]; C=Chougrane: red beds [19]; T=Taztot: trachyandesites [19]; D=Djebel Tarhat: red beds [6]; A=Série d'Abadla (Table 2). Localities are shown in Fig. 6.

The south paleopoles are plotted in Fig. 7. They are in general agreement with that from the Série d'Abadla. The magnetizations point to the south pole, and are reversed.

The distribution of paleopoles is elongated in a NNW-SSW direction. None of the rock units are very accurately dated. For the Moroccan studies an Autunian (early Permian) age is favoured in the originals [6,18,19]. An Early Permian age is also the most probable age for the Série d'Abadla, although possible stratigraphic limits are wider (see above). The spread of paleopoles of Fig. 7 could therefore reflect age differences. It is noteworthy that unit T (Taztot) is considered to be the base of the Permian sequence [20] and therefore the oldest, so the expected trend of paleopoles with age would be from south to north if the effect is caused by age differences. However, many workers from general analysis of Permian and Triassic paleopoles from Gondwana have shown that the trend with time is in the opposite sense (see below). There are no firm grounds for believing that age differences are responsible for the latitudinal spread of paleopoles.

In Table 3 the paleolatitudinal displacements (Δ) and rotations (Ψ) of the localities on the Moroccan Meseta relative to the Saharan craton (the autochthonous Série d'Abadla) are set out. In Fig. 6, for convenience of calculation , paleolatitudes are given by sampling locality rather than for each of the paleopoles of Fig. 7. None of the relative paleolatitudinal displacements are significant at $P = 0.05$. This does not mean that relative paleolatitudinal displacements have not occurred. The errors in the estimates (Table 3) are of the order of 500 km, and smaller displacements would not be detectable with the present density of observations. The results from sedimentary rocks (entries 1 and 4 of Table 3) and a volcanic unit (entry 5) show no evidence of significant relative rotation, it is only two volcanic units (entries 2 and 3) that do so. In the studies of these volcanic units, samples have been taken from 1 and 2 sites only. It is probable therefore that the secular variation has not been averaged out. Also it is difficult to determine attitudes in massive flows, and the "apparent" rotations of Table 3 could have arisen in part at least from errors in the estimates of paleohorizontal. The absence of evidence for relative rotations from sedimentary rocks strengthens this argument because in them bedding can be readily observed.

Some workers believe that paleomagnetic data from Jurassic intrusive and extrusive igneous rocks indicate clockwise rotation of the Moroccan Meseta relative to the rest of Africa [1]. Others do not consider the effect significant [2–6]. Recent detailed studies show definitively that post-Early Jurassic motions have been essentially negligible [31]. Table 3 shows that, taking into account possible errors, significant rotations do not appear to be present in data from latest Carboniferous and Permian rocks.

We conclude that there is no paleomagnetic

TABLE 3

Displacements and rotations of the Moroccan Meseta relative to the North Saharan craton

Unit	λ_o (°)	λ_e (°)	Δ (°)	ψ_o (°)	ψ_e (°)	Ψ (°)
1. Chougrane *S*, sa (33.0, 6.3)	3±2	2±3	1±3	−53±5	−53±5	−0±7
2. Chougrane *V*, si (33.0, 6.3)	3±4	2±3	1±5	−41±9	−53±5	−12±10
3. Taztot *V*, si (33.2, 6.0)	3±2	2±3	1±3	−44±5	−52±5	−9±7
4. Djebel Tarhat *S*, sa (33.5, 6.5)	−2±4	1±3	3±5	−59±8	−53±5	6±10
5. Mechra ben Abou *V*, si (32.7, 7.8)	1	1	0	−54	−54	0

λ_o and λ_e are the observed and expected paleolatitudes the latter calculated from the paleopole for the Série d'Abadla (Table 2); Δ is the relative paleolatitudinal displacement— none are significant; ψ_o and ψ_e are the observed and expected rotation and Ψ is the relative rotation (negative values indicating clockwise rotation). The calculations have been carried out using the relationships set out in Irving [21]). In the first column ($S(V)$ indicates sedimentary (volcanic) rocks, sa (si) that samples (sites) have been given unit weight in the analysis have been studied, and the values in brackets are the north latitude and west longitude in degrees of the sampling localities. Errors are at $P=0.05$. No realistic errors can be given to entry 5 (see table 2 of Westphal et al. [18]). Reference to originals is given in the caption of Fig. 7.

evidence of large relative paleolatitudinal displacements between the Moroccan Meseta and the Saharan craton since the Permian (Early Permian?). Movements of up to about 500 km are not precluded. The very low northerly paleolatitudes observed in Morocco and the very low southerly paleolatitude observed from northern Algeria suggest that the paleoequator ran between them roughly in the manner shown in Fig. 6. Paleopoles from redbeds show no significant relative rotation (paleopoles A, D and C of Fig. 7). Paleopoles from igneous rocks indicate relative rotations that are marginally significant in two instances (entry 2 and 3 of Table 3) and are not significant in the other (entry 5 of Table 3). The marginally significant differences could be caused by incomplete averaging of the secular variation or by uncertainties in the application of dip corrections in massive igneous units.

5. Comparison with paleopoles from North America and Europe

Because of the agreement between results from Morocco and the Saharan craton, it is reasonable

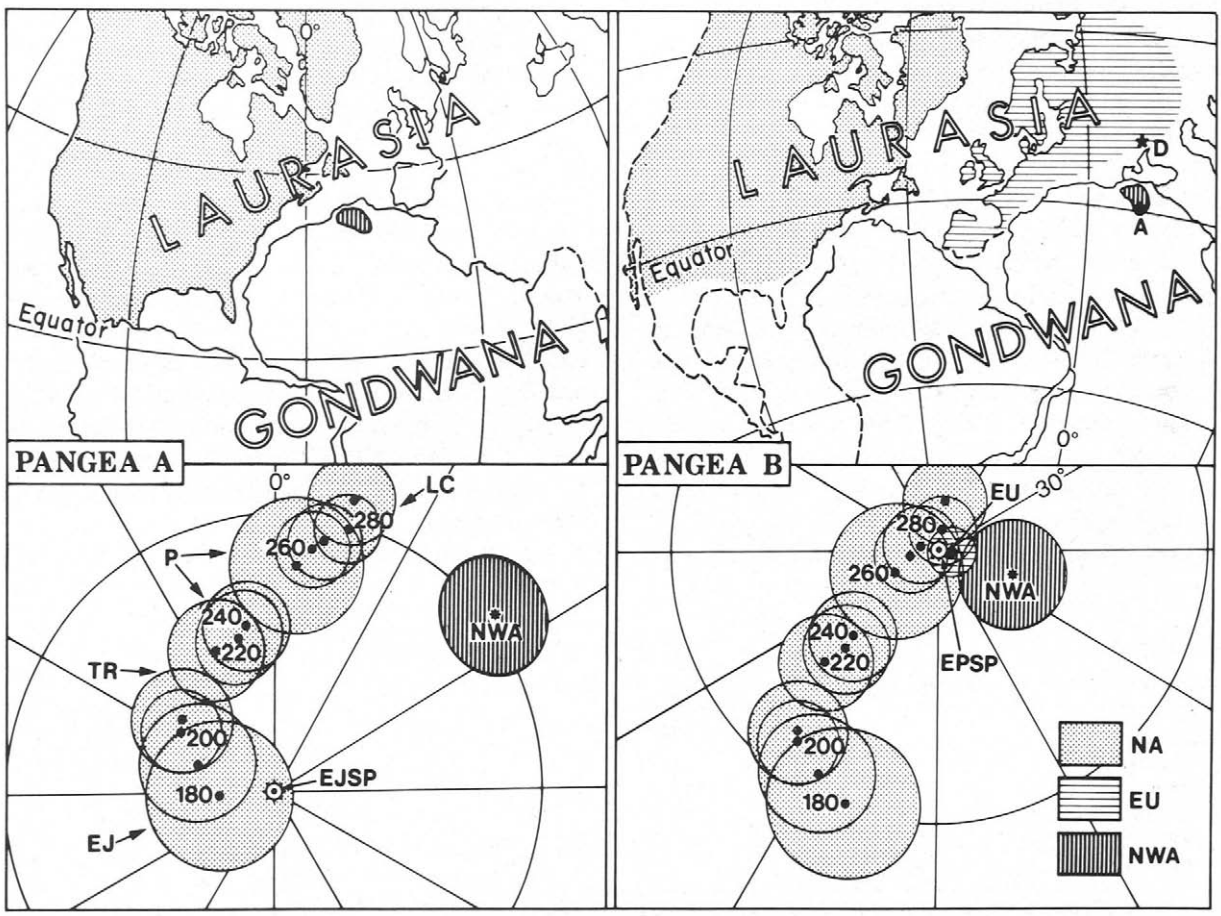


Fig. 8. Comparison of results from Nortwest Africa (NWA), North America (NA) and Europe (EU). The mean paleopoles for North America and the paleopole for Europe at 280 m.y. are documented in Morel and Irving [7]. They have been calculated by using a moving window of 30 m.y. duration. For example the mean paleopole for 280 m.y. is obtained from data in the interval 295 to 265 m.y. The geological ages of the path are indicated as follows: *LC*=latest Carboniferous (305–280); *P*=Permian (280–230); *TR*=Triassic (230–195); *EJ*=Early Jurassic (195–165). Pangea *A* is that given in Morel and Irving [7] and is very similar to that of Le Pichon et al. [27]. Pangea *B* is given in Morel and Irving [7]. *D*=Donetz Basin; *A*=Abadla.

to calculate an average paleopole for northwest Africa (NWA). The mean of the paleopoles in Fig. 6 is 32°S, 60°E ($K = 159$, $A_{95} = 6°$). This calculation has been made using Fisher's statistics which give a circular error zone. This is an imperfect procedure because the distribution is oval and it would be more accurate to use Bingham statistics [30]. The error zone should be elongated in the NNE-SSW direction, but this would not alter our result because the elongation is perpendicular to differences noted below. In Fig. 8 the NWA paleopole is compared with the North American polar wander path (APW) for Middle Jurassic to Late Carboniferous times using two different reconstructions as frames-of-reference. The first is Pangea *A* which is essentially the reconstruction of the peri-Atlantic continents given by Wegener and quantified by later workers (see caption to Fig. 8 for details). Pangea *a* is now generally accepted to be correct for the Early Jurassic [7,22–25]. The NWA paleopole does not agree with any part of the APW path for North America when reconstructed on Wegener's Pangea. On the right-hand-side of Fig. 8 the paleopoles are plotted relative to a second frame-of-reference. This is Pangea *B* [23,25,7], which is thought to pertain to the latest Carboniferous and Early Permian [7]. The NWA paleopole is now in much better agreement with the Late Carboniferous and Early Permian mean paleopoles for North America. It also agrees with the 280 m.y. (Carboniferous/Permian boundary) mean paleopole for Europe [7]. These agreements among paleopoles, when plotted in the Pangea *B* frame-of-reference, are good but are not perfect. For example the paleoequator of Pangea *B* for NWA is further south by 5° than the data for NWA would themselves indicate. It is difficult to say whether the difference is or is not significant. It should arise from imperfections in the Pangea *B* reconstruction, in the paleomagnetic record, or in the method of analysis as referred to above. The most important fact however is the poor agreement between the paleomagnetic data from northwest Africa and Pangea *A* compared to the very much better agreement with Pangea *B*. This cannot be explained by uncertainties in dating the rocks from which the northwest African paleopoles have been obtained; no age assignments within the widest possible range of uncertainty permitted by stratigraphic relationships (namely Late Carboniferous to Jurassic, see above) cam explain the discordance with Pangea *A*.

One further point of interest is that if the Pangea *B* reconstruction is correct, northwest Africa was then immediately south of the Donetz Basin (Fig. 8). The paleolatitude for the Donetz Basin obtained from beds of Early Permian age is $10 \pm 4°$N (entries 7-204, 7-308, 7-309, 7-319, 7-140 of Irving et al. [26]). That of Abadla is $6 \pm 3°$S. There is good agreement with Pangea *B*.

More data are required, but the results for northwestern Africa as they are known at present are more consistent with placing this region south of Europe in the latest Carboniferous and Early Permian (Pangea *B*) than with placing it south of North America, as is done in the classical configuration of Wegener. The later it would seem applies only for the early part of the Jurassic [7,23].

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